# Can Tabular Foundation Marginals Form a Process? Exact Projective Lifts and a 35-Dataset Stress Test

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## Abstract

Tabular foundation models provide strong one-row predictive distributions, but many decisions concern a set of future rows: portfolio returns, average claim counts, contrasts, or signed totals. Answering such queries requires cross-row dependence and, more fundamentally, one compatible stochastic process rather than unrelated finite-batch distributions. We give an exact, marginal-preserving Gaussian lift of a frozen TabICLv2 regressor. A rowwise low-rank head learns correlation from frozen hidden states; positive semidefiniteness, permutation equivariance, restriction consistency, and closure under every linear aggregate hold by construction. A precommitted stress test covers all 35 OpenML-CTR23 tasks, 630 episodes, three context sizes, six aggregate families, current tabular foundation models, exact Gaussian processes, Bayesian linear regression, and a bootstrap CatBoost process. The central performance hypothesis is rejected. Although the mean Gaussian-NLL effect is 0.5267 nat and superficially favorable, only 20/35 datasets improve ($p = 0.097$), the CRPS advantage is -0.00062, and most of the mean NLL is caused by one unstable dataset. The integrity audit also uncovers a broader trap: the usual batched TFM interface changes a row's representation when other queries are removed (maximum discrepancy 0.0711), so a PSD matrix per batch is not a projective process. Singleton evaluation restores the theorem exactly, at substantial cost. These results establish a clean construction and benchmark, but also a sharp negative boundary: coherent cross-row uncertainty does not reliably emerge from strong marginal TFM representations through a small post-hoc covariance head.


## 1 Introduction

In-context tabular foundation models (TFMs) have moved the accuracy frontier for ordinary regression and classification (Hollmann et al., 2025; Qu et al., 2026; Grinsztajn et al., 2026). Their public regression interfaces return a distribution for each query row. This is enough to estimate $\mathbb{E}[Y_i]$ or an interval for one $Y_i$, but not the uncertainty of $\sum_i a_i Y_i$: marginal variances leave every $\mathrm{Cov}(Y_i, Y_j)$ unidentified. Treating rows as independent is not innocuous. Positive dependence raises the risk of totals and averages; negative dependence can raise the risk of contrasts; either can reverse a decision.

A tempting repair is to extract query representations, learn a PSD correlation matrix, and call the result a stochastic process. Two different questions are hidden in this proposal. Validity: do predictions for overlapping query sets agree under restriction and permutation? utility: do the learned off-diagonals improve proper scores for unseen aggregate queries? Positive semidefiniteness answers neither question by itself. The first requires Kolmogorov consistency; the second is an empirical claim about dependence, not point accuracy.

We separate these questions in a deliberately controlled construction. ProjTabICL keeps every TabICLv2 mean and marginal variance fixed and learns only a correlation kernel from frozen, singleton-query representations. Its joint Gaussian can answer any linear query analytically. The identical diagonal model is therefore a strict mechanism control: any aggregate-score difference comes only from off-diagonal covariance. Hyperparameters are selected on six non-CTR23 development datasets before all 35 CTR23 tasks are scored.

The result is scientifically useful precisely because it resists a simple success narrative. A large favorable mean-NLL difference is driven by an extreme failure of the diagonal model on one dataset; the dataset win rate, randomization test, CRPS, and mechanism controls reject a broad covariance benefit. Moreover, our initial audit found that the ordinary batched backbone violates the prerequisite rowwise interface. Re-evaluating each row alone fixes projectivity without changing this negative conclusion. The main contributions are:

- a marginal-preserving projective lift of frozen TFM distributions, with proofs of PSD, restriction consistency, linear-query closure, and perturbation/proper-score consequences;
- an aggregate-query benchmark spanning six signed and unsigned query families, 35 curated regression tasks, strong joint and marginal baselines, and dataset-level inference;
- an audit showing why "PSD for every query batch" is weaker than a stochastic process, together with a singleton construction that satisfies the missing premise; and
- a frozen-protocol negative result delimiting post-hoc covariance heads: exact coherence is achieved, but the claimed performance breadth is not.

## 2 Problem and relation to prior work

Row-indexed predictive laws. Fix a context table $\mathcal{C} = \{(x_j, y_j)\}_{j=1}^{n}$. For a finite ordered query $\mathcal{Q} = (x_1, \ldots, x_q)$, a model returns a law $P_{\mathcal{C},\mathcal{Q}}$ on $\mathbb{R}^q$. A row-observation process must be permutation equivariant and projective: for every subset $S$, the marginal of $P_{\mathcal{C},\mathcal{Q}}$ on $S$ equals $P_{\mathcal{C},\mathcal{Q}_S}$. Kolmogorov's extension theorem then connects the finite-dimensional family to a process (Kallenberg, 2002). We index noisy row observations; two duplicate feature rows share a latent signal but retain independent nugget noise.

TFMs and current accuracy baselines. TabPFN introduced prior-fitted in-context tabular prediction, followed by larger and faster systems including TabICL, TabDPT, TabPFN-2.5, TabICLv2, TabDPT-Turbo, and TabPFN-3 (Qu et al., 2025; Ma et al., 2025; Grinsztajn et al., 2025; Hosseinzadeh et al., 2026; Grinsztajn et al., 2026). TabArena emphasizes that validation, tuning, and ensembling choices can change tabular rankings (Erickson et al., 2025). We therefore freeze one strong backbone, separate development from evaluation tasks, and compare dependence at identical marginals. Newer TFMs enter the point and diagonal-marginal tables; they are not mislabeled as joint predictors.

Processes, joint tables, and consistency. Gaussian processes provide coherent finite marginals by construction (Rasmussen & Williams, 2006). Neural processes amortize maps from context sets to predictive processes (Garnelo et al., 2018); Gaussian Neural Processes model correlations and establish approximation results (Bruinsma et al., 2021); Neural Diffusion Processes encode process exchangeability in richer non-Gaussian models (Dutordoir et al., 2023). MOSES enforces marginalization consistency for irregular time-series flows (Yalavarthi et al., 2026), while JoLT defines autoregressive joint tabular predictions (Shysheya et al., 2025). Most adjacent to our motivation, Klötergens et al. (2026) test compatibility across columns and targets and find violations for every evaluated TFM. Our axis is different: one target across an arbitrary set of rows, with aggregate-query scoring and an exact lift that preserves a modern TFM's one-row distributions. We do not claim that Gaussian correlations or process consistency themselves are new.

# 3 A marginal-preserving projective lift

## 3.1 Construction

Let a frozen TFM, fitted to $\mathcal{C}$, expose singleton functions

$$m_{\mathcal{C}}(x), \qquad s_{\mathcal{C}}(x) > 0, \qquad h_{\mathcal{C}}^{(e)}(x) \in \mathbb{R}^d, \quad e = 1, \dots, E. \tag{1}$$

The word singleton is essential: each quantity may depend on $(\mathcal{C}, x)$ but not on other members of $\mathcal{Q}$. A shared rowwise head $g_\theta : \mathbb{R}^d \to \mathbb{R}^r$ is normalized so $\|g_\theta(h)\|_2 = 1$. Define

$$k_{\mathcal{C}}(x, x') = \frac{1}{E} \sum_{e=1}^{E} g_\theta(h_{\mathcal{C}}^{(e)}(x))^\top g_\theta(h_{\mathcal{C}}^{(e)}(x')), \tag{2}$$

$$[\Sigma_{\mathcal{C},\mathcal{Q}}]_{ij} = s_{\mathcal{C}}(x_i) s_{\mathcal{C}}(x_j) \left[(1 - \rho_n)\mathbf{1}\{i = j\} + \rho_n k_{\mathcal{C}}(x_i, x_j)\right], \tag{3}$$

where $0 \le \rho_n < 1$ depends only on context size. Finally,

$$Y_{\mathcal{Q}} \mid \mathcal{C} \sim \mathcal{N}(m_{\mathcal{C},\mathcal{Q}}, \Sigma_{\mathcal{C},\mathcal{Q}}). \tag{4}$$

In implementation, $E = 8$ frozen TabICLv2 ensemble views, $r = 8$, and the head is a 64-unit MLP. Only this head and three $\rho_n$ values are trained.

Theorem 1 (Exact finite-dimensional consistency). Suppose the functions in equation equation 1 are deterministic rowwise functions after fitting $\mathcal{C}$. Equations equation 2– equation 4 define symmetric PSD Gaussian laws that (i) preserve $m_{\mathcal{C}}(x_i)$ and $s_{\mathcal{C}}(x_i)^2$ exactly, (ii) are equivariant to query permutation, and (iii) restrict to the same law on every sub-query. Consequently they are the finite-dimensional distributions of a conditional row-observation process.

The proof (Appendix A) writes the correlation as an identity plus a Gram matrix; restriction selects a principal submatrix. The same construction has the latent representation

$$Y_i = m_i + s_i \left[ \sqrt{\rho_n / E} \sum_e g_\theta(h_i^{(e)})^\top z_e + \sqrt{1 - \rho_n}\, \epsilon_i \right], \tag{5}$$

with independent standard Gaussian $z_e$ and $\epsilon_i$.

Corollary 1 (Arbitrary linear queries). For any matrix $A$, $AY_{\mathcal{Q}} \mid \mathcal{C}$ is Gaussian with mean $Am_{\mathcal{C},\mathcal{Q}}$ and covariance $A\Sigma_{\mathcal{C},\mathcal{Q}}A^\top$. Thus every scalar aggregate $a^TY$ is answered by $a^Tm$ and $a^T\Sigma a$ without retraining.

The variance family is not cosmetic. Polarization recovers every covariance from aggregate variances, for example $\mathrm{Cov}(Y_i, Y_j) = \frac{1}{2}[\mathrm{Var}(Y_i + Y_j) - \mathrm{Var}(Y_i) - \mathrm{Var}(Y_j)]$. Conversely, if $\|\widehat{\Sigma} - \Sigma\|_{\mathrm{op}} \le \varepsilon$, then every query obeys $|a^T(\widehat{\Sigma} - \Sigma)a| \le \varepsilon \|a\|_2^2$. Under a true Gaussian aggregate variance $v_*$ and a forecast variance $v$, the excess expected NLL is exactly $\frac{1}{2}[\log(v/v_*) + v_*/v - 1]$. Appendix A gives proofs and the conditions under which finite-rank row features approximate continuous correlation kernels.

## 3.2 The query-batch trap

Theorem 1 fails if $h_{\mathcal{C},\mathcal{Q}}(x_i)$ changes with co-queried rows. A PSD covariance for each $\mathcal{Q}$ can then have incompatible means, diagonals, and submatrices. Our predeclared audit caught exactly this failure in the standard batched path: restricting 48 queries to 24 changed a hidden coordinate by up to 0.0711 and a covariance entry by $9.24 \times 10^{-3}$. We therefore retain the batched result only as a failed diagnostic and refit all development/evaluation artifacts using a context-fitted backbone called on one query row at a time. This is an integrity correction, not outcome tuning: datasets, labels, coefficients, seeds, HPO grids, and gates are unchanged. The corrected audit is reported in Section 5.

Table 1: Predeclared comparison against identical TabICLv2 diagonal marginals. Positive score differences favor ProjTabICL. Inference is over 35 dataset means, not thousands of correlated query cells.

| Criterion | Estimate | Threshold | Pass |
|---|---|---|---|
| Mean NLL advantage | 0.5267 | $> 0$ | yes |
| NLL dataset wins | 20/35 | $\geq 21/35$ | no |
| Mean CRPS advantage | -0.00062 | $> 0$ | no |
| Randomization $p$ | 0.097 | $< 0.05$ | no |
| Mean/diagonal identity | 0.0 | $\leq 10^{-10}$ | yes |
| Restriction/permutation | 0.00e+00 | $\leq 10^{-5}$ | yes |

## 4 Experimental design

Frozen protocol and datasets. Before new CTR23 outcomes, we froze task IDs, official splits, query families, baselines, metrics, HPO grids, seeds, and success gates (protocol SHA-256 a19db...; full hashes in Appendix G). Evaluation uses all 35 tasks in OpenML-CTR23 (Fischer et al., 2023): official repeat 0, folds 0–2; context sizes 16, 64, and 256; and two nested context draws per fold. This gives 630 episodes. Every episode has six disjoint eight-row groups (48 test rows). Six non-CTR23 datasets select all hyperparameters. FreMTPL claim count, KDD17 stock return, and bike demand are held-out descriptive applications and never enter primary inference. Targets are scored after affine normalization by the corresponding training-pool mean and standard deviation; models receive native labels.

Aggregate queries. Within each group we generate a point, subset mean, $\ell_2$-normalized subset total, normalized pair difference, two-group contrast, random signed unit vector, nonnegative unit vector, and a rescaled signed duplicate. The six non-point, non-duplicate families are primary. Coefficients are deterministic and label-independent. We report Gaussian NLL and CRPS (Gneiting & Raftery, 2007), functional-mean squared error, 50/80/90/95% coverage, and width. Point normalized RMSE verifies backbone quality.

Controls and current baselines. The exact control TabICL-Diag has the same TabICLv2 means and diagonals and zero off-diagonals. Further mechanism controls are raw-feature RBF, untrained hidden cosine, and row-shuffled learned correlations. Joint baselines are exact RBF and Matérn-3/2 GPs, Bayesian linear regression (MacKay, 1992), and an eight-member bootstrap CatBoost process (Prokhorenkova et al., 2018). TabPFN-2.5 enters as a calibrated diagonal marginal. TabPFN-3 was predeclared as a point baseline; after the official local API was verified to expose a predictive variance, we additionally report its diagonal aggregate scores as an explicitly post-protocol secondary analysis. TabDPT-Turbo enters point prediction because its official API exposes no joint covariance. All marginal temperatures and classical grids are chosen globally on development data, never per CTR23 task. The full implementation/version matrix is in Appendix C.

Inference and gates. The dataset is the unit of analysis: first average query cells within a dataset, then weight 35 datasets equally. We report all effects, paired dataset bootstrap intervals (Efron & Tibshirani, 1994), sign tests, and a 100,000-draw paired randomization test. The predeclared covariance claim requires positive mean NLL and CRPS advantages over TabICL-Diag, at least 60% dataset NLL wins, randomization $p < 0.05$, exact marginal identity, and projectivity error below $10^{-5}$. Failure is reported rather than filtered.

## 5 Results

### 5.1 The exact lift fails the predeclared performance claim

Table 1 gives the decisive result. Mean NLL favors the lift by 0.5267 nat (dataset-bootstrap 95% CI $[0.0002, 1.5674]$), but only 20/35 datasets favor it and the paired randomization test

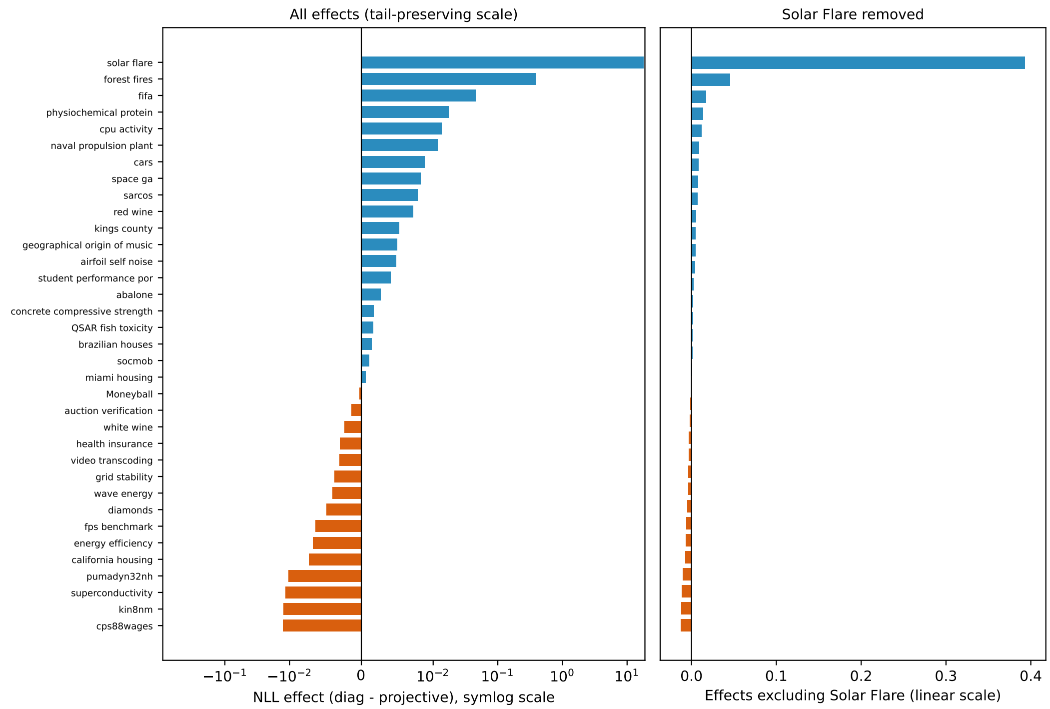


Figure 1: Dataset-balanced NLL effect of covariance at fixed TabICLv2 marginals. The left panel retains the full tail on a symmetric-log scale; the right removes Solar Flare and uses a linear scale. Positive values favor ProjTabICL. Most remaining effects are concentrated near zero.

gives $p = 0.097$. CRPS moves in the wrong direction (95% CI $[-0.00090, -0.00037]$). Only the analytic identity/integrity conditions pass; the empirical breadth claim does not. This conclusion is unchanged by singleton correction.

The NLL mean is also fragile. The largest effect is Solar Flare, where the diagonal predictive variance for some low-variation aggregates becomes catastrophically overconfident. Without that one dataset the mean advantage is 0.0133; the median is 0.00144 and the 10% trimmed mean is 0.00170. The learned head therefore provides a useful insurance effect in a rare regime, not a broad improvement. Figure 1 shows all dataset effects rather than hiding this tail.

### 5.2 Comparison with joint and current marginal baselines

Table 2 separates two findings. First, the frozen TabICLv2 mean is competitive at the point level (Appendix table 4), but its Gaussian aggregate NLL has heavy tails. Second, learning correlation from its hidden states does not reliably recover dependence: the raw-feature RBF lift and the shuffled mechanism are at least as competitive in mean NLL, while the exact GPs are more stable. Functional-mean squared error is identical between ProjTabICL and TabICL-Diag by design, so none of these variance-score differences can be attributed to better point predictions.

Figure 2 breaks the diagonal-minus-projective effect down by context size and query family. A covariance model that learned a broadly useful row geometry should improve coherent regions of this grid. Instead the effect is small and sign-heterogeneous after robust aggregation, consistent with weak dependence information in a head trained only on six external datasets.

Table 2: Primary aggregate performance, macro-averaged over 35 datasets. Lower is better except coverage (nominal 0.90). "Diag" methods assert row independence and are not joint-process baselines. Rank is more robust than raw mean NLL under catastrophic overconfidence.

| Method | NLL | CRPS | MSE | Cov.90 | NLL rank | CRPS rank |
|---|---|---|---|---|---|---|
| ProjTabICL | 7.162 | 0.3495 | 0.5697 | 0.916 | 3.11 | 4.14 |
| TabICL-Diag | 7.689 | 0.3488 | 0.5697 | 0.912 | 3.23 | 3.54 |
| TabPFN-3 (Diag) | 2.33e+05 | 0.3517 | 0.5093 | 0.886 | 4.40 | 4.03 |
| TabPFN-2.5 (Diag) | 2.33e+05 | 0.3759 | 0.5244 | 0.858 | 5.03 | 5.34 |
| GP Matérn-3/2 | 3.509 | 0.3552 | 0.5786 | 0.866 | 4.14 | 3.83 |
| GP RBF | 3.544 | 0.3606 | 0.5902 | 0.860 | 5.00 | 4.91 |
| Bootstrap CatBoost | 3.296 | 0.3701 | 0.5950 | 0.904 | 4.94 | 4.91 |
| Bayesian linear | 1672.410 | 0.3766 | 0.6060 | 0.851 | 6.14 | 5.29 |

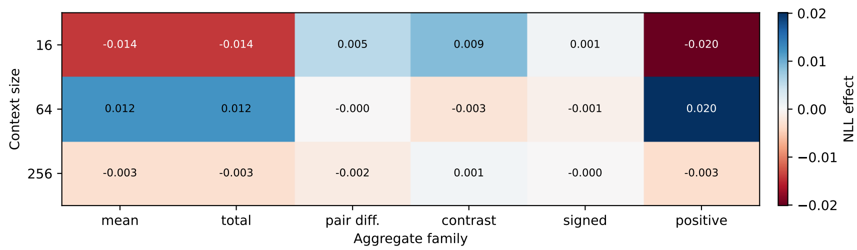


Figure 2: Median dataset NLL effect by context size and aggregate family. Positive cells favor learned covariance over the identical diagonal. The median exposes breadth without letting the Solar Flare tail set every cell.

### 5.3 Validity is exact only with a rowwise interface

The singleton model preserves every marginal diagonal to numerical zero, has minimum covariance eigenvalue above $-10^{-7}$, and passes the broad 105-episode restriction/permutation audit with maximum error 0.00e+00. The batched implementation fails by four orders of magnitude. Singleton calls cost roughly 30.6 times the backbone time for 48 queries. Thus the clean theorem exposes a practical trilemma for current APIs: batched speed, black-box reuse, and exact projectivity are not simultaneously available when preprocessing or representations depend on the query batch.

Held-out applications (Appendix D.6) reproduce the same mixed picture and are not pooled into the primary test. Earlier repository pilots had found positive projective regularization on synthetic tasks, 12 natural datasets, and three temporal datasets. We report those as prior, development-informed evidence in Appendix E; the larger frozen study supersedes them for the central performance claim.

## 6 Discussion and conclusion

The method succeeds mathematically and fails empirically. That combination clarifies what future work must change. Better marginal accuracy alone does not provide cross-row dependence; hidden similarities need not be posterior function samples; and a low-rank Gaussian head cannot represent conditional copula shape or context-specific sign changes. More promising directions are joint process pretraining, objectives that score randomly projected functions during backbone training, and non-Gaussian but analytically marginalizable families. These are materially different from adding a larger post-hoc head.

The result also changes how TFM uncertainty should be audited. Compatibility across targets or columns (Klötergens et al., 2026) and projectivity across query rows are distinct requirements. Both should be checked at the public API boundary, including preprocessing, not inferred from an attention mask or a PSD output matrix. Aggregate proper scores then test the dependence that point benchmarks cannot see.

Our limitations are explicit. The Gaussian family cannot diagnose non-Gaussian joint shape; only three context sizes and one frozen backbone are lifted; exact singleton reuse is expensive; CTR23 was previously visible to the researchers even though its labels were excluded from head selection; and three application datasets are descriptive. Within this scope, however, the evidence is unambiguous: a small post-hoc covariance head can turn TFM marginals into an exact process, but not into a reliably better aggregate forecaster.

### Reproducibility statement

The supplement contains the frozen protocol and hashes, every task ID and split rule, complete HPO grids, checkpoint/package hashes, all deviations, proofs, per-dataset results, and commands. Raw atomic episode caches include indices, coefficient hashes, timing, representations, predictions, and exceptions. The code asserts split disjointness, cell counts, marginal identity, PSD, and query restriction/permutation consistency.

### Ethics statement

The study uses public benchmark data and three previously downloaded application datasets; it involves no human-subject intervention. Regression benchmarks may encode social or historical biases and should not be interpreted as evidence that any model is appropriate for high-stakes deployment. The insurance and financial case studies are methodological illustrations, not actuarial or investment advice.

### AI use statement

OpenAI Codex was used for research-idea exploration, code drafting and debugging, experiment orchestration, literature-search assistance, proof drafting, result visualization, and manuscript drafting. The authors reviewed the generated code, ran automated and numerical integrity tests, checked cited claims against primary sources, and inspected all reported result artifacts. No AI system is listed as an author. The authors take responsibility for the final text, claims, proofs, code, and artifacts produced with AI assistance.

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# A Proofs

## A.1 Indexing convention and latent construction

The nugget in equation equation 3 indexes noisy row observations, not a deterministic function whose value must be identical at duplicate features. Formally, take an index set $\mathcal{T} = \mathcal{X} \times \mathcal{U}$, where $u \in \mathcal{U}$ is a unique observation identifier. Write $t = (x, u)$ and define

$$\kappa_{\mathcal{C}}(t, t') = (1 - \rho_n)\mathbf{1}\{t = t'\} + \rho_n k_{\mathcal{C}}(x, x'). \tag{6}$$

Two separate future observations with the same covariates share the latent term but have independent observation noise. When the scientific target is a noise-free deterministic function, the nugget should instead be modeled as measurement noise outside the latent process.

Let $\phi_e(x) = g_\theta(h_{\mathcal{C}}^{(e)}(x))$ and let $z_e \overset{\text{iid}}{\sim} \mathcal{N}(0, I_r)$ and $\epsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$, all mutually independent. Define

$$Y_t = m_{\mathcal{C}}(x) + s_{\mathcal{C}}(x) \left\{ \sqrt{\frac{\rho_n}{E}} \sum_{e=1}^{E} \phi_e(x)^T z_e + \sqrt{1 - \rho_n}\epsilon_t \right\}. \tag{7}$$

This random-variable construction is also a direct proof that all finite covariance matrices are jointly compatible.

## A.2 Proof of Theorem 1

Proof. For a query of $q$ distinct observation indices, let $\Phi_e \in \mathbb{R}^{q \times r}$ contain row $\phi_e(x_i)^T$ and let $D = \operatorname{diag}(s_{\mathcal{C}}(x_1), \ldots, s_{\mathcal{C}}(x_q))$. Then

$$\Sigma_{\mathcal{C},\mathcal{Q}} = D \left[ (1 - \rho_n) I_q + \frac{\rho_n}{E} \sum_{e=1}^{E} \Phi_e \Phi_e^T \right] D. \tag{8}$$

Every summand in brackets is PSD and $0 \le \rho_n < 1$, so the bracket and its congruence by $D$ are PSD (indeed positive definite when all $s_i > 0$ and $\rho_n < 1$). Because every $\|\phi_e(x_i)\|_2 = 1$, the bracket has unit diagonal. Hence $\Sigma_{ii} = s_{\mathcal{C}}(x_i)^2$, and the mean is fixed by definition.

Let $P$ be a permutation matrix. Rowwise evaluation gives $m_{P\mathcal{Q}} = P m_{\mathcal{Q}}$, $D_{P\mathcal{Q}} = P D_{\mathcal{Q}} P^T$, and $\Phi_{e,P\mathcal{Q}} = P \Phi_{e,\mathcal{Q}}$. Therefore $\Sigma_{P\mathcal{Q}} = P \Sigma_{\mathcal{Q}} P^T$, which is permutation equivariance. For a sub-query selected by a coordinate-selection matrix $R$, rowwise evaluation similarly gives $m_{\mathcal{Q}_S} = R m_{\mathcal{Q}}$ and $\Sigma_{\mathcal{Q}_S} = R \Sigma_{\mathcal{Q}} R^T$. A Gaussian marginal has precisely this mean and covariance, so the laws agree under restriction. The finite-dimensional family is therefore consistent. Equation equation 7 constructs the process directly; equivalently, Kolmogorov extension applies. $\square$

The premise cannot be weakened to "a PSD matrix is returned for every query." Consider means $m_{(x_1)}(x_1) = 0$ and $m_{(x_1,x_2)}(x_1) = 1$ with identity covariance for both query sets. Both laws are valid Gaussians, but the first coordinate of the second cannot marginalize to the first law. Query-dependent hidden features produce the same failure through covariance submatrices even when means happen to agree.

## A.3 Linear closure and identifiability

Proof of Corollary 1. The characteristic function of a Gaussian vector $Y \sim \mathcal{N}(m, \Sigma)$ is $\exp(iu^T m - \frac{1}{2}u^T \Sigma u)$. Substituting $A^T u$ gives the characteristic function of $AY$ as $\exp(iu^T Am - \frac{1}{2}u^T A\Sigma A^T u)$, proving the claim. $\square$

Proposition 1 (Polarization identifies covariance). If the variance $V(a) = \mathrm{Var}(a^T Y)$ is known for every $a \in \mathbb{R}^q$, then the covariance matrix is unique. In particular,

$$\Sigma_{ij} = \tfrac{1}{2}\{V(e_i + e_j) - V(e_i) - V(e_j)\} = \tfrac{1}{4}\{V(e_i + e_j) - V(e_i - e_j)\}. \tag{9}$$

Proof. Expand the quadratic form $V(a) = a^T \Sigma a$ at the displayed vectors. The diagonal follows from $V(e_i)$ and every off-diagonal follows from the first identity; hence no two covariance matrices induce the same variance oracle. □

This explains the benchmark design. Point queries identify only the diagonal. Subset totals and positive dense queries emphasize positive covariance; differences, contrasts, and signed dense queries expose the sign and geometry of off-diagonal entries. The scaled duplicate is excluded from the primary average and used only to audit the quadratic scaling identity.

### A.4 Perturbation and proper-score consequences

Proposition 2 (Uniform functional-variance bound). For symmetric matrices $\widehat{\Sigma}, \Sigma$ and any $a$,

$$|a^T(\widehat{\Sigma} - \Sigma)a| \le \|\widehat{\Sigma} - \Sigma\|_{\mathrm{op}} \|a\|_2^2. \tag{10}$$

Consequently a covariance operator error $\varepsilon$ controls all unit-norm linear-query variances simultaneously.

Proof. Set $M = \widehat{\Sigma} - \Sigma$. Cauchy–Schwarz and the definition of the spectral norm give $|a^T M a| \le \|a\|_2 \|Ma\|_2 \le \|M\|_{\mathrm{op}} \|a\|_2^2$. □

Proposition 3 (Expected Gaussian-NLL regret). Suppose the true aggregate law is $Y \sim \mathcal{N}(\mu_*, v_*)$ and the forecast is $\mathcal{N}(\mu, v)$, with $v_*, v > 0$. Relative to the true forecast, expected negative-log-likelihood regret is

$$\frac{1}{2}\left[\log \frac{v}{v_*} + \frac{v_* + (\mu - \mu_*)^2}{v} - 1\right]. \tag{11}$$

At a shared correct mean this reduces to $\frac{1}{2}[\log(v/v_*) + v_*/v - 1] \ge 0$, with equality only at $v = v_*$.

Proof. The forecast NLL is $\frac{1}{2}[\log(2\pi v) + (Y - \mu)^2/v]$. Taking expectation uses $\mathbb{E}(Y - \mu)^2 = v_* + (\mu_* - \mu)^2$. Subtract the same expression with $(\mu, v) = (\mu_*, v_*)$. Nonnegativity is the KL divergence between the two univariate Gaussians. □

For our mechanism comparison, $\mu$ is exactly shared. Let $v_D = \sum_i a_i^2 s_i^2$ be the diagonal aggregate variance. The projective variance is $v_D + 2\sum_{i<j} a_i a_j \Sigma_{ij}$. Equation equation 11 makes clear why neither coherence nor PSD guarantees better scores: the learned correction must have the right sign and magnitude for each query under the data-generating law.

### A.5 Approximation scope of normalized finite-rank heads

Proposition 4 (Conditional correlation-kernel approximation). Let $\mathcal{X}$ be compact, let $h : \mathcal{X} \to \mathcal{H}$ be continuous and injective, and let $k_* : \mathcal{X}^2 \to \mathbb{R}$ be a continuous PSD kernel with $k_*(x, x) = 1$. Assume its Mercer expansion converges uniformly. If the rowwise neural class is dense in continuous vector-valued functions on $h(\mathcal{X})$, then for every $\epsilon > 0$ there is a rank $r$ and a unit-normalized rowwise map $g$ such that

$$\sup_{x,x'} |g(h(x))^T g(h(x')) - k_*(x, x')| < \epsilon. \tag{12}$$

Proof. Uniformly truncate the Mercer feature map to $\psi_r(x) = (\sqrt{\lambda_j} e_j(x))_{j=1}^r$. Uniform kernel convergence and $k_*(x, x) = 1$ imply $\|\psi_r(x)\|_2^2$ is uniformly close to one for sufficiently large $r$. Thus $\bar{\psi}_r(x) = \psi_r(x)/\|\psi_r(x)\|_2$ is continuous and its inner product uniformly approximates $k_*$. Injectivity on compact $\mathcal{X}$ makes $h$ a homeomorphism onto its image. Universal approximation supplies a network uniformly close to $\bar{\psi}_r \circ h^{-1}$; normalization is continuous away from zero, so its normalized output preserves the desired error after choosing the approximation tolerance sufficiently small. □

This is a model-class statement, not a claim about our fixed rank-eight head or about the sufficiency of TabICLv2 representations. Gaussian Neural Processes provide broader process-level approximation theory (Bruinsma et al., 2021). Our negative experiment indicates that the conditional premises needed for a useful low-rank post-hoc approximation are not reliably realized in practice.

# B Full experimental protocol

## B.1 Episode construction and leakage controls

For each official CTR23 split, context rows are sampled only from the training fold and all 48 query rows only from the test fold. Contexts are nested across 16, 64, and 256 rows within each replicate. Query indices are unique within an episode and groups are disjoint. The normalization mean and scale are computed from the complete training fold solely for metric comparability; each estimator receives native labels. Query coefficients are generated from a stable hash of dataset, split, and replicate and never access query labels.

For a group $G$ of eight coordinates, the families are:

1. a point $e_i$;
2. a subset mean $|S|^{-1}\mathbf{1}_S$;
3. an $\ell_2$-normalized subset total $|S|^{-1/2}\mathbf{1}_S$;
4. a pair difference $(e_i - e_j)/\sqrt{2}$;
5. a two-subset contrast normalized to unit $\ell_2$ norm;
6. a random signed unit vector;
7. a random nonnegative unit vector; and
8. a positive rescaling of family 6 for identity auditing.

Families 2–7 are the frozen primary set. Each family is evaluated on six groups, but these 36 cells are averaged before a dataset enters inference.

## B.2 Dataset inventory

Development datasets are Pol, Elevators, Sulfur, Ailerons, House-16H, and Fried, with three deterministic 70/30 splits and three context draws per split (162 episodes). The application panel contains 120,000 FreMTPL rows, 122,749 KDD17 stock-return rows, and 17,379 bike-demand rows, using the same random-split design (81 episodes). The panel is semantically useful but too small for confirmatory inference.

# C Hyperparameter optimization and implementations

## C.1 Proposed head

Marginal temperatures are selected first using development point-query NLL over $\{0.25, 0.5, 0.75, 1, 1.5, 2, 3, 4, 6, 8\}$. The selected values for context sizes 16, 64, and 256 are 4, 1, and 1. We then perform leave-one-development- dataset-out HPO over rank $\{8, 16, 32\}$, hidden width $\{0, 64\}$, weight decay $\{0, 10^{-4}, 10^{-3}\}$, and learning rate $10^{-3}$. AdamW uses at most 400 epochs, patience 40, and gradient norm clipping at 5. The selected head is rank 8, hidden width 64, weight decay $10^{-4}$, trained for 13 epochs. Three final seeds produce

Table 3: All 35 evaluation datasets. Rows are the full task size; each episode nevertheless uses at most 256 context and 48 query observations.

| Dataset | Rows | $p$ | Dataset | Rows | $p$ |
|---|---|---|---|---|---|
| Moneyball | 1,232 | 14 | health_insurance | 22,272 | 11 |
| QSAR_fish_toxicity | 908 | 6 | kin8nm | 8,192 | 8 |
| abalone | 4,177 | 8 | kings_county | 21,613 | 21 |
| airfoil_self_noise | 1,503 | 5 | miami_housing | 13,932 | 15 |
| auction_verification | 2,043 | 7 | naval_propulsion_plant | 11,934 | 14 |
| brazilian_houses | 10,692 | 9 | physiochemical_protein | 45,730 | 9 |
| california_housing | 20,640 | 8 | pumadyn32nh | 8,192 | 32 |
| cars | 804 | 17 | red_wine | 1,599 | 11 |
| concrete_compressive_strength | 1,030 | 8 | sarcos | 48,933 | 21 |
| cps88wages | 28,155 | 6 | socmob | 1,156 | 5 |
| cpu_activity | 8,192 | 21 | solar_flare | 1,066 | 10 |
| diamonds | 53,940 | 9 | space_ga | 3,107 | 6 |
| energy_efficiency | 768 | 8 | student_performance_por | 649 | 30 |
| fifa | 19,178 | 28 | superconductivity | 21,263 | 81 |
| forest_fires | 517 | 12 | video_transcoding | 68,784 | 18 |
| fps_benchmark | 24,624 | 43 | wave_energy | 72,000 | 48 |
| geographical_origin_of_music | 1,059 | 116 | white_wine | 4,898 | 11 |
| grid_stability | 10,000 | 12 | | | |

$\rho_{16} = 0.1201 \pm 0.00035$, $\rho_{64} = 0.1198 \pm 0.00050$, and $\rho_{256} = 0.1184 \pm 0.00021$ (standard deviations across three fits).

TabICLv2 is frozen at checkpoint tabicl-regressor-v2-20260212.ckpt, eight ensemble views, and 199 quantiles used to estimate each marginal variance. The standard batched cache was discarded from primary inference after the audit. Singleton inference fits the context/preprocessors once and invokes prediction separately for each query row; weights are not reloaded between rows.

### C.2 Baselines

- TabPFN-2.5: package 6.3.0, pinned local default-regressor checkpoint, eight estimators. Global development temperatures are selected with the same grid as TabICLv2.
- TabPFN-3: package 8.5.0 in an isolated environment, pinned official default v3 regressor checkpoint, eight estimators with automatic feature-coverage scaling disabled. Its marginal temperature is selected on the same six development datasets. The frozen protocol allowed this model as a point baseline if official local weights became accessible. Once we verified that the API also exposes predictive variance, we added the diagonal aggregate analysis post-protocol, labeled it secondary, and did not use it in any primary gate or model choice.
- TabDPT-Turbo: package 1.2.0 in an isolated environment, official tabdpt1_2.safetensors, eight prediction ensembles. Its public API exposes a mean but no predictive covariance, so only nRMSE is reported; uncertainty values are never fabricated.
- Exact GPs: RBF and Matérn-3/2 kernels with development length-scale multiplier in $\{0.5, 1, 2\}$, context-standardized inputs, and learned/noise-stabilized Gaussian posterior covariance.
- Bayesian linear: conjugate/evidence-estimated ridge posterior over context-standardized features.
- CatBoost process: eight context bootstrap members; their prediction covariance plus out-of-bag residual diagonal noise. Development grid: iterations $\{200, 500\}$, depth $\{4, 6\}$, learning rate $\{0.03, 0.1\}$. The selected settings are $(500, 4, 0.03)$ at 16 rows and $(500, 4, 0.1)$ at 64/256 rows.

Classical preprocessing uses context-only category codes, median imputation, and standardization. Native TFM preprocessing is retained except for documented package edge cases in Section F. All baselines predict the identical cached context/query indices.

# D Additional results

## D.1 Point prediction

Table 4: Point prediction macro-averaged over 35 datasets. TabDPT-Turbo is mean-only, so distributional entries are intentionally blank.

| Method | nRMSE | NLL | CRPS | Cov.90 |
|---|---|---|---|---|
| TabICLv2 | 0.7200 | 24.680 | 0.3441 | 0.918 |
| TabPFN-3 | 0.6883 | 3.29e+05 | 0.3476 | 0.903 |
| TabPFN-2.5 | 0.7045 | 3.29e+05 | 0.3693 | 0.876 |
| TabDPT-Turbo 1.2 | 0.7133 | – | – | – |
| GP Matérn-3/2 | 0.7472 | 8.544 | 0.3615 | 0.884 |
| GP RBF | 0.7579 | 10.169 | 0.3676 | 0.877 |
| Bootstrap CatBoost | 0.7569 | 4.158 | 0.3787 | 0.923 |
| Bayesian linear | 0.7723 | 1971.078 | 0.3894 | 0.872 |

## D.2 Fixed-marginal mechanism controls

Table 5: Correlation mechanism controls with the same TabICLv2 means and marginal variances. Positive $\Delta$NLL favors the row-correlation model over the exact diagonal. The shuffled, hidden-cosine, and raw-feature controls are PSD and preserve the diagonal but do not use the trained head as intended.

| Method | NLL | $\Delta$NLL vs. diag. | CRPS | Cov.90 |
|---|---|---|---|---|
| ProjTabICL | 7.162 | 0.5267 | 0.3495 | 0.916 |
| TabICL-Diag | 7.689 | 0.0000 | 0.3488 | 0.912 |
| Row-shuffled head | 7.125 | 0.5631 | 0.3496 | 0.916 |
| Untrained hidden cosine | 7.164 | 0.5243 | 0.3499 | 0.913 |
| Raw-feature RBF lift | 6.749 | 0.9396 | 0.3500 | 0.914 |

## D.3 Wall-clock cost

Table 6: Per-episode wall-clock prediction time on the evaluation panel. The TabICLv2 row reports one context fit plus 48 singleton calls; timings include model prediction but not cache serialization or scoring. Hardware was two NVIDIA H100 GPUs for neural models and CPU for classical baselines.

| Method | Median (s) | Mean (s) | 90th pct. (s) |
|---|---|---|---|
| TabICLv2 singleton | 1.638 | 1.750 | 2.165 |
| TabPFN-3 | 0.687 | 0.692 | 0.740 |
| TabPFN-2.5 | 0.304 | 0.317 | 0.350 |
| TabDPT-Turbo 1.2 | 0.200 | 0.201 | 0.203 |
| GP Matérn-3/2 | 0.015 | 0.041 | 0.110 |
| GP RBF | 0.013 | 0.037 | 0.090 |
| Bootstrap CatBoost | 0.588 | 1.146 | 2.041 |
| Bayesian linear | 0.001 | 0.001 | 0.002 |

### D.4 All dataset effects

The effect is diagonal NLL/CRPS minus projective NLL/CRPS; positive favors learned covariance at exactly shared marginals.

Table 7: All 35 fixed-marginal covariance effects.

| Dataset | NLL effect | CRPS effect |
|---|---|---|
| solar_flare | 17.98300 | 0.00000 |
| forest_fires | 0.39337 | -0.00036 |
| fifa | 0.04531 | -0.00126 |
| physiochemical_protein | 0.01722 | -0.00001 |
| cpu_activity | 0.01352 | -0.00028 |
| naval_propulsion_plant | 0.01172 | 0.00067 |
| cars | 0.00883 | -0.00029 |
| space_ga | 0.00823 | 0.00017 |
| sarcos | 0.00784 | -0.00045 |
| red_wine | 0.00720 | -0.00051 |
| kings_county | 0.00523 | -0.00004 |
| geographical_origin_of_music | 0.00498 | -0.00034 |
| airfoil_self_noise | 0.00481 | 0.00009 |
| student_performance_por | 0.00406 | -0.00150 |
| abalone | 0.00270 | -0.00025 |
| concrete_compressive_strength | 0.00174 | -0.00056 |
| QSAR_fish_toxicity | 0.00167 | -0.00018 |
| brazilian_houses | 0.00144 | -0.00007 |
| socmob | 0.00107 | 0.00005 |
| miami_housing | 0.00057 | -0.00004 |
| Moneyball | -0.00022 | -0.00055 |
| auction_verification | -0.00139 | -0.00033 |
| white_wine | -0.00235 | -0.00194 |
| health_insurance | -0.00298 | -0.00042 |
| video_transcoding | -0.00306 | -0.00011 |
| grid_stability | -0.00371 | -0.00100 |
| wave_energy | -0.00404 | -0.00169 |
| diamonds | -0.00485 | 0.00002 |
| fps_benchmark | -0.00635 | -0.00114 |
| energy_efficiency | -0.00672 | 0.00012 |
| california_housing | -0.00730 | -0.00079 |
| pumadyn32nh | -0.01042 | -0.00285 |
| superconductivity | -0.01157 | -0.00191 |
| kin8nm | -0.01227 | -0.00208 |
| cps88wages | -0.01255 | -0.00201 |

### D.5 Query-set consistency audit

Table 8: Maximum normalized discrepancies over 35 datasets and three context sizes (one fold-0, replicate-0 episode each; 105 episodes).

| Query mode | Mean | Variance | Hidden | Covariance |
|---|---|---|---|---|
| Batched | 5.91e-03 | 9.24e-03 | 7.11e-02 | 9.24e-03 |
| Singleton | 0.00e+00 | 0.00e+00 | 0.00e+00 | 0.00e+00 |

For each episode we compare (i) a 48-row query with a deterministic 24-row restriction and (ii) the same 48 rows under a random permutation. We audit mean, calibrated variance, hidden representation, and final covariance after realigning coordinates. The singleton path is expected to agree up to floating point because every row call is identical after the deterministic context fit. The audit is deliberately broader than a unit test and covers heterogeneous/categorical/missing-value tasks.

### D.6 Application panel

Table 9: Held-out application results. These three datasets do not enter the 35-dataset primary estimate or its hypothesis tests.

| | NLL | | | CRPS | | |
|---|---|---|---|---|---|---|
| Dataset | Proj. | Diag. | Effect | Proj. | Diag. | Effect |
| bike_sharing | 0.859 | 0.853 | -0.0062 | 0.3552 | 0.3546 | -0.00054 |
| fremtpl_claim_count | 298.919 | 338.263 | 39.3443 | 0.5188 | 0.5182 | -0.00061 |
| kdd17_stock_return | 1.396 | 1.385 | -0.0108 | 0.5852 | 0.5836 | -0.00158 |

FreMTPL aggregates resemble claim-count portfolio questions; KDD17 aggregates resemble baskets and long–short contrasts; bike-demand aggregates resemble fleet totals and temporal contrasts. The sampled rows are exchangeable under our tabular episode construction; we do not assert a deployment-grade temporal forecasting protocol.

## E Earlier experiments and how they informed this study

The repository contains a sequence of progressively stronger projectivity pilots. They motivated the present protocol but are not independent confirmation:

- A semisynthetic static neural-process model beat a much larger direct scalar-query network on all 120 held-out dense/scaled cells, with 1.774 nat mean NLL advantage and identity residual below $9 \times 10^{-8}$.
- A six-dataset zero-shot natural-label anchor exposed calibration mismatch: aggregate NLL improved, but natural point NLL worsened and the breadth condition failed.
- A corrected 12-dataset natural-scale escalation gave a 0.346 mean NLL advantage over a direct query model. At fixed marginals, learned covariance improved NLL by only 0.004 and CRPS by 0.00176; this modest mechanism signal motivated the stronger frozen-head control used here.
- On Jena Weather, Electricity, and Traffic, temporal projective models generalized compositionally better than direct-query networks, but low-rank covariance failed a later component gate and Electricity remained a clear failure against TACTiS.

The present study follows the precommitted recommendation from those pilots: use a strong modern marginal backbone, natural tasks, broad query families, strong joint baselines, fixed-marginal ablation, and dataset-level inference. Its negative gates therefore supersede the earlier go/no-go signal for the specific post-hoc-head claim.

## F Protocol deviations and edge cases

All deviations were logged without replacing a task, query, seed, or outcome:

1. Four 16-shot Solar Flare CatBoost bootstrap members had constant labels. CatBoost rejects this degenerate fit, so the mathematical solution (the observed constant function) replaces only that member.
2. Twelve FPS Benchmark TabPFN-2.5 episodes had a feature entirely missing in context. Such columns are deterministically dropped from context and query. One constant-label Solar Flare episode required matching logits to the distribution-border dtype before computing variance.

3. Some 16-shot bike-demand contexts made a categorical feature constant, triggering a TabPFN-2.5 validation error. On that exact error only, native categories are replaced with context-derived integer codes; unseen query categories map to $-1$. No query label is used.
4. The batched projectivity audit failed. The method definition was corrected to singleton queries, the head was retrained, and all primary predictions were rerun. Both the failed batched artifacts and corrected artifacts are retained. This change enforces a stated integrity gate; it changes no performance gate or evaluation data.
5. TabPFN-3 was frozen as an optional point-only comparison. The acquired official API exposed full marginal distributions, so we also temperature- calibrated its variance on the development panel and report a diagonal aggregate analysis. This is a post-protocol secondary baseline: it does not enter the ProjTabICL-versus-TabICL-Diag claim, and no primary choice was changed after seeing it.

# G Reproduction and artifact map

The machine-readable protocol is config.json. Its SHA-256 is 4e74...; the prose protocol hash is a19db.... The cache root is external to the source tree to avoid filling the repository. Core commands, from the experiment directory, are:

```
python extract_tabicl.py --stage dev  --query-mode singleton
python train_head.py --query-mode singleton
python extract_tabicl.py --stage eval --query-mode singleton
python score_projective.py --query-mode singleton
python extract_classical.py --stage eval
python extract_tabpfn.py --stage eval --device cuda
tabpfn3_env/bin/python extract_tabpfn3.py --stage eval --device cuda
tabdpt_env/bin/python extract_tabdpt.py --stage eval --device cuda
python score_baselines.py --query-mode singleton
python audit_query_projectivity.py --query-mode singleton
python score_applications.py
python generate_paper_artifacts.py
```

Dataset shards can be split with –shard-index and –num-shards; writes are atomic and valid cached outputs are skipped. Every episode stores context/query indices, their SHA-256 hashes, metric normalization, query coefficients and hash, means, variances, hidden states where applicable, elapsed time, package/checkpoint metadata, and the complete split identity. Result tables are regenerated from Parquet/JSON artifacts, not copied from terminal output.

The final integrity checklist verifies 630 evaluation episodes, 81 application episodes, finite outputs, official train/test disjointness, 48 unique query rows per episode, coefficient determinism, classical/projective PSD, exact projective diagonals, complete baseline caches, query permutation/restriction, and compilation of this manuscript. The failed batched audit lives under results/projectivity_audit_batched; it is never silently overwritten.